MILITARY SPECIFICATION

READER, MICROHEMATOCRIT, SPIRAL TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of circular microcapillary tube reader which indicates hematocrits directly in percent red cells and is commonly known as a spiral reader.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Shipping, Fiberboard

Military

MIL-P-116	Preservation-packaging, Methods of
MIL-M-7298	Manual, Technical, Commercial Equipment

STANDARDS

Federal

FED-STD-66	Steel, Chemical Composition and Hardenability

FSC 6630

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Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Base. The base shall be fabricated from cast metal and the feet shall be rubber.

3.1.2 Disks. Disks shall be fabricated from corrosion-resisting steel, 300 series of FED-STD-66, not less than 0.045 inch thick.

3.1.3 Capillary tube holder. The tube holder shall be fabricated of clear, rigid plastic.

3.2 Design and construction. Reader shall consist of a base, reading disks and a capillary tube holder, joined to form a single unit. Reading disks shall be secured to the top center of the base by means of a pin or bolt which shall permit free rotation of the disks without binding. The disks shall be capable of rotation with respect to each other by means of finger holds, and after setting shall maintain their relative positions. One disk shall be capable of unlimited rotation and the other disk shall rotate within the limits of its stop (see 3.2.2). The capillary tube holder shall be securely fixed to one side of the base and to the center pin or bolt.

3.2.1 Base. The base shall be circular. The bottom surface shall have at least three equally spaced feet. The reader shall be stable on a flat, level surface.

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3.2.2 Bottom disk. The bottom disk shall have an engraved scale suitable for direct reading of percent red cells. Scale values shall increase in a clockwise direction around the circumference of the disk. The length of the scale shall be a nominal 24 inches. The scale shall be marked with graduation lines for all integers from 2 to 100. From 2 to 20 on the scale each graduation line shall be numbered. Over 20, every fifth graduation line shall be numbered. The lower portion of the scale shall be expanded so that 65 ± 2 hundredths of the scale length is below the 20 mark. The disk shall be furnished with a stop which aligns the hundred mark with the index line of the tube holder (see 3.2.4).

3.2.3 Top disk. The top surface of the top disk shall be marked with clear and legible instructions for use engraved on its surface and a spiral curve extending to the perimeter of the disk. The spiral curve shall be clearly visible through the tube holder.

3.2.4 Capillary tube holder. A longitudinal groove for accommodating capillary tubes shall be located in the top surface of the holder. The groove shall accommodate capillary tubes of the following dimensions:

Outside diameter 1 through 7 millimeters (mm.)

Length 13 to 75 mm.

An index line on the holder shall align with the 100 mark on the scale when the holder is at the top of the hematocrit scale and against the stop.

3.3 Dimensions. The disks and base shall have the following diameters:

Bottom disk	8-1/8 ± 1/8 inches (in.)
Top disk	7-1/8 ± 1/8 in.
Base	7 ± 1/8 in.

3.4 Finish. Outer surface of the base shall be enameled. The disks and holder shall be smoothly finished, with burrs, roughness and sharp edges removed.

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3.5 Marking. All markings shall be permanent, clear and legible. The instructions for use, the scale markings, and the spiral curve (3.2.3) shall be engraved.

3.5.1 Identification marking. Each reader shall be permanently marked with the name or registered trademark of the manufacturer.

3.6 Performance. With the specified capillary tubes in place in the tube holder, rotation of the disks in accordance with the manufacturer's instructions shall permit alignment of the spiral curve with the serum-air interface and the red-white cell interface of centrifuged blood-filled capillary tubes and permit accurate reading of the hematocrit (percent red cells). When tested as specified in 4.4.1, hematocrit determinations on the reader and calculations (in percent) from the measurements on the standard tubes in the tube holder shall not differ from each other by more than 3 percent.

3.7 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the contractor shall furnish 15 copies of the approved technical manual to the contracting officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the contracting officer. Waiver of submission of 15 additional copies shall be at the option of the Government.

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3.8 Workmanship. Workmanship shall be first class throughout. Readers shall be free from defects listed herein and other defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of quality. Certificate of quality, supplied by the manufacturer of the corrosion-resisting steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number (NSN), the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table I. Unit of product shall be one reader.

Table I. Sampling for examination

Examination	Inspection level	AQL (percent defective)
Visual examination		
Major defects	II	1.0
Minor defects	II	2.5
Dmensional examination	S-2	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II. Unit of product shall be the reader, and acceptance number shall be zero for all sample sizes.

Table II. Sampling for tests

Characteristic	Requirement	Test procedure	Inspection level
Accuracy	3.6	4.4.1	S-3

4.3 Examination. The reader shall be examined to determine compliance with requirements of this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with Table III but shall not be restricted to the defects delineated therein.

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Table III. Classification of defects

Category	Defects
Major	
101	Component missing.
102	Disks not able to be rotated readily.
103	Disks or holder not secured to base.
104	Markings not complete, correct, legible or permanent.
Minor	
201	Reader not stable or free of wobbling when placed on a flat, level surface.
202	Reader not free of sharp edges.
203	Reader not free of burrs or pits.
204	Tube holder not free of cracks.

4.3.2 Dimensional examination. The reader shall be examined for defects in dimensions. Any dimension not within the specified tolerances shall be classified as a defect. In order to determine acceptability of tube holder groove dimension, tubes of specified dimensions shall be placed in tube holder (3.2.4).

4.3.3 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

Table IV. Examination of preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.

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Table IV. Examination of preparation for delivery (cont'd)

Examination	Defect
Workmanship	Inadequate application of components such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Accuracy. A tube approximately 75 mm. in length by 1.5 to 7.0 mm in diameter shall be marked so as to represent a capillary tube which is being used for the determination of percent blood cells (hematocrit) in blood. Marks shall be made to represent bottom of red blood cells, percentage of red cells and serum air or plasma air interface. Marks shall be measured with an instrument of known accuracy. Any tolerance on measuring instrument shall be subtracted from the tolerance specified in 3.6. Distance between marks for bottom of red cells and serum air interface shall be 50 mm. (nominal). Hematocrit marks shall be so placed as to represent between 40 and 50 percent blood cells. Instructions on face of reader shall be followed, and hypothetical percent of blood cells of the marked tube shall be determined.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit package. Each reader shall be packaged in accordance with MIL-P-116, method IC-2.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

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2 Packing. Packing shall be Level A, B or C as specified
2).

2.1 Level A. Four (4) readers shall be packed in a box
d for a Type 2 load and constructed in accordance with
85, Class 3, Style 3; PPP-B-601, overseas type; PPP-B-621,
; or PPP-B-636, Class weather-resistant, Grade V3c. Closure
apping shall be as specified in the appendix of the
ble box specification.

2.2 Level B. Four (4) readers shall be packed in a box
d for a Type 2 load and constructed in accordance with
85, Class 1, Style 3; PPP-B-601, domestic type; PPP-B-621,
; or PPP-B-636, Class domestic. Closure shall be as
ed in the appendix of the applicable box specification.

2.3 Level C. The subject commodity shall be packed in
ial commercial containers of the type, size and kind
used for the purpose, so constructed as to insure
ce and safe delivery by common or other carriers, at
st rate, to point of delivery called for in the contract
ase order.

Marking.

Interior packages. Each unit package shall be marked
dance with MIL-STD-129. When labels are utilized,
ofing shall be required only when applicable carton is
ed of water-resistant material. Contract or purchase
nber and name of contractor shall be shown.

.2 Exterior container. Exterior container shall be marked
dance with MIL-STD-129.

NOTES

Intended use. The reader covered by this specification
led to be used to make precision readings of hematocrits
e (large quantity) analyses, as well as in research

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6.2 Ordering data. Procurement documents should specify:

a. Title, number and date of this specification.
b. National Stock Number.
c. Applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification is intended to cover the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6630-00-585-1378	READER, MICROHEMATOCRIT, SPIRAL TYPE

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DSA-DM

Project No. 6630-0238
cs

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-R-37198 READER, MICROHEMATOCRIT, SPIRAL TYPE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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